

Performance For All Editions on ODA X6-2S/M

The Oracle Database Appliance is not new. The idea comes from early 2002 with the 'Row Iron' plan: an appliance with Sun server and Oracle Database.

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The idea was resurrected in 2011 with ODA V1 and has evolved through capacity and flexibility to X3-2, X4-2 and X5-2. This year the new one, X6-2 has been announced with some major changes:

- Multiple models: X6-2S, X6-2M and the previous X5 is renamed 'HA'
- Single server models (2S and 2M) whereas previous ODAs were all 2-nodes clusters
- Ability to run Standard Edition (previous models were only for Enterprise Edition)

This article describes the 'Small' and 'Medium' models that are out at the time of writing. A 'Large' one will extend the range of those ODA 'Lite' soon.

Standard Edition

The previous Oracle Database Appliances were built for Enterprise Edition. The vision was to have a server that is easy to manage, and requires minimal administration staff. The goal was also optimized costs with Capacity on Demand where you buy licences only for the cores activated. For those goals, Enterprise Edition comes immediately to mind. Automation is easier with Enterprise Edition features and some options, and Capacity on Demand makes sense only when licensing on a number of cores. So the first idea was that ODA was made only for Enterprise Edition.

Of course, it was possible to licence only the minimum cores for Enterprise Edition and install Standard Edition in a guest VM of a virtualised ODA. But administration is not easy and performance is very bad (guest VMs do not have direct access to disks).

Today, customers want an appliance for Standard Edition as well. And this has reached another level when Standard Edition One (SE1) has been abandoned and Standard Edition Two (SE2) came with new limits.

Some medium companies have consolidated everything on VMware. They have only a couple of Oracle Databases but lots of other VMs in a large ESX farm. And unfortunately they have to buy Oracle Database licences for the whole. With the least expensive SE1, counting all sockets in ESX servers was still affordable. For small companies with limited users, licensing in named user metric instead of processor was possible as the minimum for SE was 5 NUP. But SE2 changed those minimums, that are now higher

and proportional to the total number of servers, and is more expensive than SE1.

So you can't consolidate Oracle databases in your VMWare cluster for licensing reasons. You probably don't want to manage several hypervisors. The only solution is to dedicate a server (and maybe even the storage). However, when your goal is consolidation, you don't want to setup and maintain a physical server with separate storage just to host a few small Oracle databases. This is where an appliance looks like the right solution, but a 2-nodes cluster with Enterprise Edition, which is what ODA was up to X5-2, is too expensive for just a few small databases without critical HA requirement.

Today, the ODA X6-2S and ODA X6-2M can be installed in Standard Edition. Note that you cannot mix editions: all your ORACLE_HOMES will be either Standard or Enterprise. It's a choice you do at ODA installation.

It's a single server with one socket (X6-2S) or two sockets (X6-2M), so if you are in Standard Edition you will have to license 1 processor for X6-2S or 2 processors for X6-2M. If you are in NUP, the minimum is 10 NUPs per ODA and you can even run 11g with your SE1 licences where the minimum is 5 NUP in total. If you are in Enterprise Edition in core metric, you can go with Capacity on Demand where you can buy licences one by one: activating cores 2 by 2 (because core factor is 0.5). The ODA is the easiest solution from a licensing point of view because it's the only platform where Oracle accepts deactivating cores in BIOS as a valid hard partitioning solution, according that it is done with the 'configure-core-count' script provided with ODA. Of course, I'm talking about small on-premises solutions.

Capacity on Demand is also possible on Exadata and Oracle Cloud Machine, but that's not in the scope of activating a number of cores that you can count on your fingers.

Standby Database

You need a standby database. Even with the clustered ODAs where RAC ensures the High Availability of the instance, there is no protection of data better than backups, though backups can take a long time to restore and recover. RAC does not protect the database from media failure or block corruption because the storage is shared and when the ODA is patched as a whole, RAC does not help to minimise maintenance windows with rolling upgrades. For this reason, most ODA installations come with 2 or 3 ODAs – a pair in Data Guard for production, and non-critical databases can run on the standby site or on a third ODA. You realise then why the ODA up to X5-2 was not considered as a cost effective solution for a few small databases, given that you had to buy at least two clusters (i.e 4 servers with 36 cores each) and then need to licence the activated cores with Enterprise

Edition - Capacity on Demand being not so flexible because you need to keep symmetry between the two nodes, and probably between the two sites.

ODA X6 changes everything there. You can buy two X6-2S and you have highly redundant hardware on two sites for an affordable cost.

If you have Enterprise Edition licences, then you will setup Data Guard. If you are in Standard Edition, you can have a good availability with Dbvisit standby, of course, you don't need to and you can build your own scripts to manage the manual standby. But if you go to an appliance with lots of automation and reliability, you probably want to build the HA on the same philosophy. With Dbvisit standby, you can reduce both the Recovery Point Objective (the archive log lag) and the Recovery Time Objective to a few minutes (but with manual intervention - nothing like FSFO here). To create the standby database on an ODA X6 you can use the 'odacli create-database -io' to create the instance only, so that it is registered by the ODA, and then Dbvisit standby will re-create the instance and do the duplicate.

Another solution, when there is only one datacenter, is to have the primary databases hosted by one on-premises ODA, and run the standby databases on a public cloud service.

Be careful on the cost when you switchover because the cost of download data transfer may be higher than upload.

Performance – CPU

I'm talking about entry-level, cost effective, Standard Edition, small databases. But don't get it wrong. Those appliances have amazing performance for a database workload.

The ODA X6-2S processor is an Intel Xeon E5-2630 v4 running at 2.20GHz. One socket, 10 cores, 20 threads. This is perfect to run a Standard Edition 2 database which is caged at 16 threads. And, believe me, you can run a large OLTP application on 16 threads. If you don't, there's probably some tuning to do before blaming the hardware. Remember that with Standard Edition 2 you can use instance caging so that you can lockdown some non-critical databases and keep resources available for the critical one. Note that databases created in ODA have `cpu_count` set but no resource plan (except for maintenance window) so the recommendation is to set one.

The ODA X6-2M has two sockets, which gives 20 threads. The processor frequency is not very high and you should care about it if you migrate a single-threaded application from a 3GHz server. But even at 2.2GHz ODA X6 has very good processor.

Load Profile	Per Second	Per Transaction	Per Exec	Per Call
DB Time(s):	39.9	545.6	0.00	25.32
DB CPU(s):	38.8	530.8	0.00	24.64
Logical read (blocks):	18,520,084.4	253,310,980.1		
... Instance Efficiency Percentages (Target 100%)				
Buffer Nowait %:	100.00	Redo NoWait %:	100.00	
Buffer Hit %:	100.00	In-memory Sort %:	100.00	
... Top 10 Foreground Events by Total Wait Time				
Event	Waits	Total Wait Time (sec)	Wait Avg (ms)	% DB Wait time Class
DB CPU		11.7K		97.3

FIGURE 1

Figure 1 is my test with cached SLOB (<https://kevinclosson.net/slob>) with 40 threads running.

So with high buffer cache hits, (which is the goal of OLTP) running the ODA X6-2M at full capacity gives us 18 million logical reads per second. Think about it. If your OLTP reads 1000 blocks on average for a transaction, you can handle 18000 transactions per second. As a comparison, consider that a properly tuned query, with correct indexes, should not read more than 10 or 20 blocks to get few rows. If it's not the case, then show me the execution plan...

Even a single ODA X6-2S can handle quite a load from one or few databases. The database provisioning interface (command line with 'odacli create-database' or GUI with the Web Console) gives the choice of predefined configuration with 'shapes' (sets the cpu_count, memory, and related parameters) and 'classes' ('OLTP', 'DSS', 'In-Memory'). Note that with Standard Edition you can set only 'OLTP' but this is just an indication that data warehouses and BI workloads often need Enterprise Edition features and options, no limitation here.

Performance - Storage

ODA X6 is full NVMe SSD, which means you get rid of the CPU overhead to transform PCIe to SAS protocol, and this gives optimum access to flash disks. But once again, let's test it with SLOB (this time with no buffer cache hits) in order to get real life results. Figure 2 shows 10 threads doing single block reads.

This is 80,000 IOPS which is not bad at all. It is just an example. We can go further. I got 800,000 IOPS with the same 10 sessions workload using prefetching, keeping average 'db file parallel read' below 1ms. But look at this 80,000 IOPS numbers above: 100% of blocks were read in less than a millisecond. Actually, from the `v$wait_event_histogram_micro`, I checked that 99% of the reads were faster than 256 microseconds and 97% faster than 128 us. This makes the gap between disk and RAM very small with a few terabytes of disk available depending on ODA X6 model and ASM redundancy (only 'normal' redundancy for 2S and 2M without additional storage as there are only 2 NVMe disks).

I did another test on a similar workload where 100% of the data read is also updated. I reached the rate of 70MB/s of redo

Load Profile	Per Second	Per Transaction	Per Exec	Per Call					
~~~~~									
DB Time(s):	10.0	300.2	0.03	20.99					
DB CPU(s):	2.4	72.3	0.01	5.06					
Background CPU(s):	0.0	0.3	0.00	0.00					
Logical read (blocks):	79,829.9	2,405,105.7							
Block changes:	32.2	968.6							
Physical read (blocks):	79,144.1	2,384,445.5							
Physical write (blocks):	2.6	79.4							
Read IO requests:	79,143.0	2,384,411.3							
Write IO requests:	1.4	41.4							
Read IO (MB):	618.3	18,628.5							
Write IO (MB):	0.0	0.6							
...									
Instance Efficiency Percentages (Target 100%)									
~~~~~									
Buffer Nowait %:	100.00	Redo NoWait %:	100.00						
Buffer Hit %:	0.86	In-memory Sort %:	100.00						
...									
Top 10 Foreground Events by Total Wait Time									
~~~~~									
Event	Waits	Total Wait Time (sec)	Wait Avg(ms)	% DB Wait time Class					
-----									
db file sequential read	23,843,794	2679.3	0.11	89.2 User I/O					
DB CPU		723.4		24.1					
...									
~~~~~									
% of Waits									

Event	Total Waits	<1ms	<2ms	<4ms	<8ms	<16ms	<32ms	<=1s	>1s

db file sequential read	23.7M	100.0	.0	.0	.0	.0	.0	.0	

FIGURE 2

generated (13000 transactions per second) while keeping 95% of commit time (log file sync) lower than 4 milliseconds.

This is a good performance acceptable for critical databases, OLTP or BI.

ACFS

On ODA X6 you have the choice to create a database directly on the ASM diskgroups, or have an ACFS mount point for them. The default is ACFS (with one filesystem per database for data, and a common filesystem for Fast Recovery Area). However, the performance is not the same. I wasn't able to get more than 80,000 IOPS on ACFS.

Only once the datafiles moved to +DATA we were able to go further. This problem is known and we can expect that it will be solved.

Whatever you choose at creation, you can easily move datafiles in 12c Enterprise Edition as it is an online operation. Unfortunately, because of diskgroup compatibility, you cannot put 11g datafiles directly on ASM.

You probably don't want to do snapshots in production, so put it in +DATA if you want the best performance. Maybe in development, especially in multitenant, you will use snapshot clone for thin provisioning. This is a case where you may want to use ACFS. In Standard Edition, you cannot use ACFS snapshots for the database, so better create it directly in +DATA and get the best performance.

Configuration

When you install the ODA, you choose the edition (standard or enterprise). The versions available in both cases are 11.2.0.4

and 12.1.0.2. The ODA X6-2S/M run only 'bare metal': no virtualization, which makes it simple. If you want to install some applications, you will have to install them on the same machine as the database. This is not a recommendation but can be interesting for other Oracle products licensed on cores.

You can leave all CPU cores activated, or activate on a multiple of two. Once you use Capacity on Demand, activating fewer cores to save on Enterprise Edition processors licences, you will be able to increase them later (but no way back). For the storage you choose at ODA installation the partitioning between +DATA and +RECO will mainly depend on the expected backup size that you will keep on the ODA. There is more flexibility here because in addition to the predefined external or internal backup configuration where +DATA is 80% or 40%, you can customise with any value between 10% and 90%.

One final recommendation: don't forget that all the simplicity and reliability comes from standardised components that have been configured, tested and tuned to run together.

Don't try to customise it further than allowed. Do not try to run other versions than the supported ones (currently 11.2.0.4 and 12.1.0.2) and keep in mind that the goal is to run databases that do not need additional one-off patches, so that ODA patching is kept easy.

In summary, those two ODA X6 2S and 2M extends the Oracle Database Appliance philosophy to a higher level: easier to setup, addressed at all customer sizes, and more and more performance. If you considered the ODA as a small Exadata, then you can look at ODA X6-2M and 2S as even smaller ones, but still focused on performance.



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